

AI-Powered Predictive Maintenance for Industrial Equipment

Project Plan

Timeline · Milestones · Resource Allocation

Team Members

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1. Project Overview

This document outlines the project plan for the AI-Powered Predictive Maintenance system. The project spans January to mid-May 2025 and is divided into three major phases: Data Analysis & Preprocessing, Model Development, and MLOps & Deployment.

2. Phase Timeline & Gantt Chart

The table below summarizes the timeline for all major tasks across the project phases.

Task	Duration	Responsible Members	Done
Data Collection & Exploration	Jan 1 – Jan 20	Alaa, Rowan, Nada, Taghreed	✓
CMAPSS Data Preprocessing	Jan 10 – Feb 10	Alaa, Taghreed, Nada	✓
MIMII Data Preprocessing	Jan 10 – Feb 10	Alaa	✓
Advanced Analysis & Feature Engineering	Jan 20 – Feb 15	Alaa, Taghreed	✓
RUL Prediction (LSTM) – Build & Train	Feb 15 – Mar 31	Rowan, Alaa	✓
Acoustic Anomaly Detection (Autoencoder) – Build & Train	Feb 15 – Mar 31	Alaa	✓
Model Finetuning & Optimization	Apr 1 – Apr 15	Alaa, Rowan, Nada, Taghreed	✓
MLflow Integration	Apr 16 – May 10	Alaa, Nada	✓
Model Deployment (FastAPI)	Apr 16 – May 15	Alaa	✓
Monitoring & Continuous Improvement	May 10 – May 15	Nada	✓

3. Milestones

#	Target Date	Milestone	Success Criteria
M1	Feb 15, 2025	Data Analysis & Preprocessing Complete	All datasets cleaned, preprocessed, features engineered
M2	Apr 15, 2025	Models Trained & Optimized	LSTM (RUL) and Autoencoder (Anomaly) models finalized
M3	May 10, 2025	MLflow Integration Complete	Experiments tracked, model registry populated
M4	May 15, 2025	Deployment Live	FastAPI endpoints serving predictions, monitoring active

4. Resource Allocation

Team Member	Primary Responsibilities	Phase Focus
Alaa Mohamed	CMAPSS & MIMII preprocessing; LSTM & Autoencoder models; All deployments; MLflow	All Phases
Rowan Nasser	Data exploration & visualization; RUL prediction model co-development	Phase 1 & 2
Nada Ahmed	CMAPSS preprocessing; MLflow integration; Monitoring pipeline	Phase 1 & 3
Taghreed Mohammed	CMAPSS preprocessing & advanced analysis; Feature engineering	Phase 1 & 2